

Problem: Solve the equation:

$$2x^2 + x = 2$$

Solution: This is a quadratic equation of the general form

$$ax^2 + bx + c = 0$$

To see this just subtract 2 from both sides

$$2x^2 + x - 2 = 0$$

So we have

$$a = 2$$

$$b = 1$$

$$c = -2$$

From the formula sheet, this equation has at most two solutions:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

which means that the two solutions

x_1 and x_2 are

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$$x_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

Since we have

$$a = 2$$

$$b = 1$$

$$c = -2$$

We get the two solutions:

$$x_1 = \frac{-1 - \sqrt{1^2 - 4 \cdot 2 \cdot (-2)}}{2 \cdot 2} = \frac{-1 - \sqrt{17}}{4} \approx -1,28$$

$$x_2 = \frac{-1 + \sqrt{17}}{4} \approx 0,78$$